

ICRA 2027 System Flow and Development Boundary

Status at 2026-08-18: **NO_GO_P2**. The active conference-development route is the **P0 + P5 backup paper path**. P2 is frozen. Gate 0B is separately blocked by a required upstream process failure and has no P0 performance result.

This document is a development and evidence map. It does not claim certification-level proof, physical isolation, or a formal safety guarantee.

Governing sources and interpretation

This flow is governed by the conference [scope contract](#), the activation addendum in the [implementation plan](#), the Supervisor verdict in [SUPERVISOR_LOG.md](#), and the [baseline freeze manifest](#). The historical [Gate 0 qualification report](#) remains evidence, but its Gate 0B performance label is superseded by the Supervisor review because the raw log exposes a hidden required-process failure. The requested [docs/icra27/ICRA_CODE_MAP.md](#) is not present at the reviewed HEAD; the existing [CODE_MAP.md](#) and current source were used only to verify current class names, topics, and call order.

The diagram uses two kinds of statement:

- **Current implementation** names an existing call, type, or observed runtime path.
- **Scope contract** states the boundary that future ICRA work must preserve. A scope-contract node is not evidence that the behavior has already passed qualification.

The current integrity monitor is the authoritative current-state monitor within this system. P0/Predictor may read it in one direction as a current-integrity prior, but may not write back to it. P0 prediction and P2 preference remain advisory. Original EGO collision, dynamics, refinement, and feasibility logic remains authoritative for motion feasibility. P5 final and P5 runtime are separate gates and together form the only hard integrity-gate authority in the IAP layer.

End-to-end authoritative system flow

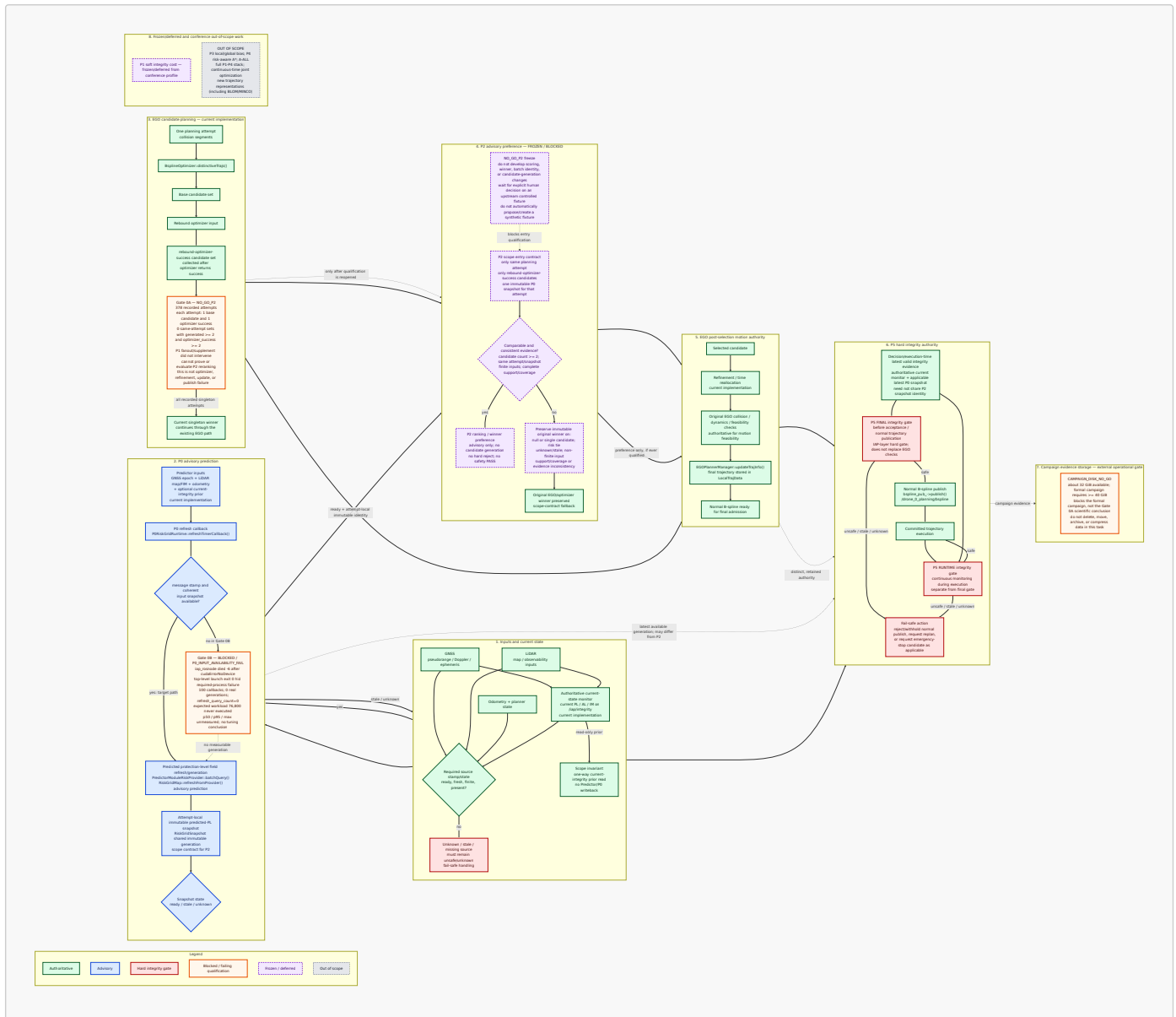


Figure interpretation: the actual current planner path feeds the rebound-optimizer-success candidate set into P2 before the selected candidate enters the original EGO refinement and feasibility stages. The figure therefore does **not** claim that every candidate passed final EGO feasibility before P2. Gate 0 showed that the selected singleton candidate still reached recorded refinement, `updateTrajInfo()`, and normal publication events in all 378 attempts; those narrow planner events support `N0_G0_P2`. They do not erase the simultaneous `iap_rosnode/integrity-validator` failure and must not be presented as full-system qualification.

P5 final and P5 runtime intentionally appear as separate hard gates. Current code reacquires the latest available risk snapshot for each P5 evaluation and consumes the current monitor independently; P5 is not required to use P2's snapshot identity. This independence does not make P5 a replacement for EGO collision or dynamics authority.

Gate 0 blockers and their meaning

Gate 0A: `N0_G0_P2` (narrow candidate-qualification verdict)

The recorded natural path produced 378 planning attempts, each with one base candidate and one rebound optimizer success. There was no attempt with both `generated >= 2` and `optimizer_success >= 2`, so there is no eligible same-attempt set on which reranking could be evaluated. P1 collision fanout and supplement were absent. The singleton candidate path remained operational through optimizer success, selection, refinement/feasibility, `updateTrajInfo()`, and normal B-spline publication.

Consequence: P2 is frozen and blocked. Reopening it requires human review and explicit acceptance of an upstream controlled fixture. This document neither recommends nor authorizes automatic synthetic-fixture creation. This verdict answers only whether the observed natural candidate path supports P2 reranking; it is not a PASS for live integrity input, P0, P5 or the complete IAP system.

Gate 0B: `BLOCKED / P0_INPUT_AVAILABILITY_FAIL`

The raw P0 log shows `iap_rosnode` loaded the GPU mapping backend on a machine with no CUDA device and died with exit `-6`. The runner observed only the top-level launch exit 0 and wrote `planner_crash=false`, so the required-process failure was hidden from the manifest. The capture then contains 100 refresh callbacks and zero successful generations: first `message_stamp_unavailable`, then `snapshot_unavailable`. These are downstream input-availability symptoms after the integrity producer died. Because

`refresh_query_count` stayed at zero, the expected 76,800-query workload did not execute and no full-grid latency percentile was measured.

The nine Gate 0A logs similarly show `iap_rosnode` exit `-6` and validator exit `2` with zero integrity messages. That failure does not create missing P2 candidates and therefore does not overturn the narrow singleton verdict, but it prevents any broader system qualification.

Consequence: this is an upstream required-process/input-availability blocker, not evidence that P0 is too slow. P0 p50, p95 and max remain unmeasured. Explicitly select the qualification CPU backend, prove a live integrity report and one real P0 generation, and only then rerun the unchanged Gate 0B workload. Worker count, ROI, horizons, and refresh period must not be tuned from this result.

CAMPAIGN_DISK_NO_GO

Available space is about 32 GiB, below the frozen formal-campaign threshold of at least 40 GiB. This independently blocks a formal campaign but does not alter the Gate 0A scientific conclusion. Storage remediation is outside this document-only task; no data may be deleted, moved, archived, or compressed here.

Current development route

- 1. Overall state remains **NO_GO_P2**.
- 2. The active route is the **P0 + P5 backup paper path**.
- 3. The unique next task is `ICRA-002 / GATE_0B`: add an explicit qualification-only `gpu|cpu` backend choice, use `cpu`, and restore a live `iap_rosnode`, valid integrity input and at least one real successful P0 generation in the fixed smoke.
- 4. Only after the smoke passes, run the fixed Gate 0B protocol once and verify the 76,800-query shape, at least 20 successful generations, and `p95 <= 400 ms`. Do not reinterpret failed callbacks as latency samples.
- 5. After Gate 0B passes, independently validate P5 final and P5 runtime hard-gate behavior for unsafe, stale, and unknown states.
- 6. Keep P2 frozen. Only a human-reviewed, explicitly accepted upstream controlled fixture may reopen P2 qualification.
- 7. Before P2 qualification is restored, do not develop P2 scoring, winner selection, batch identity, or candidate-generation modifications.
- 8. Resolve the formal-campaign disk threshold separately before any formal campaign; do not perform disk remediation as part of this documentation task.

Component status

Status vocabulary used below:

- **ACTIVE**: an in-scope path exists and is part of the current P0+P5 development route; it is not necessarily qualified.
- **PASS**: the specifically cited Gate 0 observation succeeded; this is not a certification claim.
- **BLOCKED**: progress depends on an unresolved prerequisite or external operational condition.
- **FAILED-QUALIFICATION**: the fixed qualification gate ran and did not meet its acceptance conditions.
- **FROZEN**: development is prohibited until the stated human/gate decision occurs.
- **OUT-OF-SCOPE**: excluded from the ICRA conference route.

Component/path	ICRA role	Authority	Current state	Evidence/problem	Next gate
current integrity monitor	Current PL/AL/IM and current-state integrity source	Authoritative current-state monitor within the system	ACTIVE	<code>/iap/integrity</code> is read independently by P0 and P5; no Predictor/P0 writeback is allowed	Preserve one-way separation while diagnosing P0 inputs
Predictor/current prior path	Builds advisory predictor input; optional one-way current-integrity prior	Advisory consumer of current monitor	ACTIVE	Current code has the read path, but Gate 0B did not form a coherent usable generation input	Diagnose message stamp and snapshot availability without changing decisions
P0 snapshot generation	Refreshes the predicted-PL field	Advisory	BLOCKED	<code>iap_rosnode</code> died <code>-6</code> on the GPU backend; 100 callbacks, 0 real generations, <code>refresh_query_count=0</code> ; workload and p95 unmeasured	Explicit CPU smoke; restore one real generation, then run fixed Gate 0B once
P0 immutable attempt snapshot	Supplies an attempt-local immutable predicted-PL snapshot	Advisory	BLOCKED	Immutable generation type exists, but no Gate 0B generation was available and the P2 attempt/set identity contract is not qualified	Gate 0B pass; P2 use remains frozen independently

Component/path	ICRA role	Authority	Current state	Evidence/problem	Next gate
EGO candidate generation	Produces base candidates from collision segments via <code>distinctiveTrajs()</code>	Original EGO planning authority	PASS	All 378 attempts produced one base candidate; this is operational but insufficient for P2 comparison	No P2 change; only human review may authorize an upstream controlled fixture
rebound optimizer	Optimizes base candidates and produces the rebound-optimizer-success candidate set	Original EGO optimizer authority	PASS	378 inputs produced 378 successes; singleton status is not optimizer failure	Preserve optimizer behavior on active P0+P5 route
P2 ranking	Prefers a lower predicted-risk candidate within one eligible attempt/snapshot set	Advisory only	FROZEN	No attempt had a qualifying multi-candidate success set; reranking cannot be evaluated	Explicit human acceptance of upstream controlled fixture, then new qualification
EGO refinement/collision/dynamics	Refines selected candidate and enforces motion feasibility	Authoritative for motion feasibility	PASS	All selected singleton candidates reached the recorded refinement/update/publish path; this does not claim every pre-P2 candidate was finally feasible	Preserve authority and validate regression when P0+P5 work changes nearby integration
P5 final gate	Hard integrity admission before normal trajectory publication	IAP-layer hard integrity authority	ACTIVE	Existing call is before normal B-spline publish and can use a newer snapshot than P2; backup-route independent qualification remains	After Gate 0B passes, test safe/unsafe/stale/unknown final behavior
P5 runtime gate	Continuous hard integrity monitoring of the committed trajectory	IAP-layer hard integrity authority	ACTIVE	Separate runtime evaluation path uses latest execution-time evidence	After Gate 0B passes, test safe/unsafe/stale/unknown runtime behavior independently
campaign evidence storage	Stores formal run artifacts and evidence	External operational prerequisite; no algorithm authority	BLOCKED	About 32 GiB available versus at least 40 GiB required	Resolve capacity separately before formal campaign; preserve existing data
P1 soft cost	Deferred journal/full-stack path	No ICRA conference authority	OUT-OF-SCOPE	Excluded from conference profile; source may remain frozen	None on the ICRA conference route
P3/P4/A-ALL/full stack/new trajectory work	Excluded research expansion	None in ICRA conference scope	OUT-OF-SCOPE	P3 bias, P4 risk-aware A*, A-ALL, full P1-P4, continuous-time joint optimization, and new trajectory representations are excluded	None on the ICRA conference route

Invariants for future development

1. The current monitor must never accept Predictor/P0 writeback or override. The claimed separation is logical and one-way, not unproven physical isolation.
2. P0 must provide P2 with one attempt-local immutable predicted-PL snapshot.
3. P2 may compare only rebound-optimizer-success candidates from the same planning attempt and the same immutable snapshot.
4. P2 is always advisory preference. It must never become a hard integrity gate, emit a safety PASS, or hard-reject a candidate.
5. P2 must not generate candidates. Its eligible input set must be unchanged by ranking.
6. Null, single-candidate, risk-tie, unknown/stale, non-finite, incomplete-support, and inconsistent-evidence cases must preserve the immutable original winner.
7. Original EGO collision, dynamics, refinement, and feasibility authority must remain intact.

8. P5 final and P5 runtime are separate. Together they are the IAP layer's only hard integrity-gate authority, and they do not replace EGO motion-feasibility checks.
9. P5 may use the latest valid integrity evidence at decision or execution time; it is not required to share P2 snapshot identity.
10. Unknown, stale, missing-source, invalid, and non-finite integrity evidence must fail safe and must never be interpreted as low risk.
11. Diagnostic and evidence code must not alter candidate generation, optimization, ranking, refinement, feasibility, publication, or safety action.
12. Documentation, experiments, and paper claims must not assert certification-level proof, certified active perception, physical isolation, or formal safety guarantees.

Explicit conference exclusions

The ICRA route excludes P1 soft cost, P3 local/global bias, P4 risk-aware A*, A-ALL, the full P1-P4 stack, continuous-time localization-planning joint optimization, new trajectory representations such as BLOM/MINCO, and certification-level PHMI/formal-safety claims. These items must not be pulled into the active P0+P5 route merely because their source remains in the frozen baseline.

Future ICRA development records

- Place all subsequent ICRA branch development records in [docs/icra27/dev/](#).
- Use the recommended filename form `YYYY-MM-DD_<topic>.md`.
- Every record must link back to this [ICRA_SYSTEM_FLOW.md](#).
- Every record must identify the affected flow nodes, the invariants it preserves, its test evidence, any Gate-status transition, and unresolved problems.
- A flow-node status may change only after evidence verifies the applicable gate. Implementation completion alone is not sufficient to change **BLOCKED** or **FAILED-QUALIFICATION** to **PASS**.